

Problem-Solution Fit

Date	07 July 2026
Team ID	SWTID-2026-3218
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] - Financial institutions and banking firms - Loan officers and credit analysts - Individuals seeking a new credit card	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Applicants often have limited financial disclosures - Application details may be incomplete - Income sources vary widely - Credit history on file may be minimal - Heavy reliance on manual review steps	5. AVAILABLE SOLUTIONS PROS & CONS [AS] - Manual checks: reliable but slow and expensive - Rule-based systems: quick but not adaptable - ML-based models: fast, scalable, accurate, and improve with historical data
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Manual review takes considerable time - Evaluation errors happen due to human involvement - High daily volume of incoming applications - Approval decisions are delayed and inconsistent	9. PROBLEM ROOT / CAUSE [RC] - Application volumes keep growing - Verification is still largely manual - Human-driven decisions vary in consistency - Applicant information is often incomplete - Risk evaluation lacks efficiency	7. BEHAVIOR + ITS INTENSITY [BE] - Applicants want a fast turnaround on decisions - Banks handle a very high daily application count - There's a clear need for quick, consistent, dependable evaluation

3. TRIGGERS TO ACT [TR] • A rise in the number of credit card applications • Growing demand for faster approvals • Customers expecting instant decisions • Pressure to cut down operating costs 4. EMOTIONS BEFORE / AFTER [EM] Before: Uncertainty, frustration, prolonged waits, and stress After: Quicker approvals, greater confidence, transparency, and improved trust	10. YOUR SOLUTION [SL] Build a Credit Card Approval Prediction System that leverages Machine Learning (Logistic Regression, Decision Tree, Random Forest, and XG Boost) integrated into a Flask-based web application. The system instantly predicts approval or rejection using applicant details, improving accuracy, cutting down processing time, and helping banks make more dependable decisions.	8. CHANNELS OF BEHAVIOR [CH] ONLINE • Banking websites • Mobile banking apps • Online credit card portals OFFLINE • Bank branch locations • Customer support centers • Financial advisors
--	---	---